

SECTION 5 – POWER REQUIREMENTS

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5-1 INTRODUCTION

The existing system PDU is replaced during the upgrade to ACGD. The MR upgrade includes the new Power Distribution module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3) which distributes power to most MR system components and provide the proper voltage configuration for the ACGD modules. Refer to Table 5-1 for the required customer power specifics. Refer to Section 5-2 CRITICAL POWER REQUIREMENTS for specifications of required facility input for the system.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Note

Location of the new PDU in the ACGD/PDU Cabinet relative to the existing system PDU may require replacement of the customer provided feeder wires from the MDP to the PDU. Refer to requirements and information throughout this section (Section 5 – POWER REQUIREMENTS). Local and national codes may require an electrician to provide and/or install new feeder wires to the PDU.

TABLE 5-1
REQUIRED CUSTOMER POWER

MR COMPONENT	VOLTAGE (VAC)	FREQUENCY	PHASE	MAX. AMPS	COMMENTS
Power Distribution Unit (PD1) Module located in the lower portion of ACGD/PDU Cabinet (MR3) SEE NOTE 1	480/415/400/ 380/208/200	50/60 $\pm$ 3 Hz	(3+GND) See Comments	See Note 2	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet. Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Note 1 Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable, or cord per National Electric Code (NEC) 2002 Articles 250.134, 300.3, 517.13 or 1999 Articles 250-134, 300-3, 517-13. 2 Maximum amps dependent on voltage selected. Refer to Section 5-2, CRITICAL POWER REQUIREMENTS, for configuration.					

5-2 CRITICAL POWER REQUIREMENTS

The facility shall provide a Main Disconnect Control (MDC) with Low Voltage Low Energy local and multi-point remote control capability in the feeder lines that supply input power to the Shield Cooler Cabinet (if applicable for system magnet configuration) and the Power Distribution Unit (PD1). All work is to be done in accordance with national and local electrical codes. Refer to Section 5-3, POWER DISTRIBUTION SYSTEM, for MDC capability and set up.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Configuration:

- Recommend input configuration 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet.
- Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.

Frequency: 50 ± 3 Hz or 60 ± 3 Hz

System PDU Voltage: 200/208/380/400/415/480 VAC $\pm 10\%$

Note

A second circuit breaker in the Main Disconnect is required for Shield/Cryo Cooler Cabinet. Refer to Section 5-3 Main Disconnect Control.

Shield Cooler Compressor Voltage (Water Cooled type): 380/400/415 VAC 50 Hz or 460/480 VAC 60Hz

Daily Voltage Variation: $\pm 10\%$ from nominal under worst case line and load regulation.

Voltage Transients: Maximum transient voltage above nominal waveshape not to exceed 200 V for the nominal voltages 380, 400, 415, or 480 Vrms.

Voltage Transients: Phase-to-phase voltages must be within 2% of the lowest phase-to-phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.

Phase Balance: Difference between the highest phase line-to-line voltage and the lowest phase line-to-line voltage must not exceed 2%.

5-2 CRITICAL POWER REQUIREMENTS (Continued)**Facility Ground:**

- Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger.
- Main facility ground wire to be insulated.
- Ground impedance to earth at power source to be 2 ohms or less.
- Main facility ground wire to be bonded at every distribution box in an approved grounding block.

Maximum Momentary Demand: The power demands for the system with ACGD are specified as a function of the duration of the power demand. Table 5-2 lists points on the curve.

The power system feeding the Signa system with ACGD must be designed to meet the specifications of less than 4% regulation when loaded at the 5.0 second allowable consumption. For short intervals the Signa system with ACGD power demands can exceed the 5.0 second value and the line voltage delivered to the system will sag below the 4% regulation. The Signa system with ACGD is designed to tolerate these **short** voltage sags.

TABLE 5-2
SYSTEM WITH ACGD POWER DEMANDS DETAILS

TIME INTERVAL	PDU POWER DEMAND	SHIELD COOLER COMPRESSOR POWER DEMAND	TOTAL SYSTEM POWER DEMAND
Continuous	45 KVA	9 KVA	54 KVA
5.0 seconds	~56.2 KVA	9 KVA	~65.2 KVA
1.0 seconds	~65 KVA	9 KVA	~74 KVA
0.10 seconds	~65 KVA	9 KVA	~74 KVA
0.04 seconds	~65 KVA See Note 1	9 KVA	~74 KVA
Note 1 The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA.			

Average (while scanning) Power Demand: 54 KVA consisting of 45 KVA for PDU and 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet.

Standby (no scan) Power Demand: 13.4 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU and 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet.

5-3 POWER DISTRIBUTION SYSTEM

■ 5-3-1 Main Disconnect Control (MDC)

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE MAIN DISCONNECT CONTROL FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Required Main Disconnect Control shall consist of a three-pole, 600 VAC circuit breaker trip rated for the appropriate current. The short-circuit current interrupting rating of the breaker must be sized to accommodate fault current available.

Note

The MDC circuit for the PDU shall have an undervoltage device that automatically trips the breaker when the applied coil voltage drops to a specific percent of the rated value.

Note

The MDC circuit for the Shield Cooler Compressor Cabinet shall provide auto restart upon return of normal power to maximize proper uninterrupted magnet operation and minimize cryogen consumption of the system.

WARNING!

THE MDC CIRCUIT FOR THE SHIELD COOLER COMPRESSOR CABINET AUTO RESTART FUNCTION MUST BE CONTROLLED BY THE EMERGENCY OFF FUNCTION.

■ Main Disconnect Control is to be located so the panel is visible to Power Distribution Unit (PD1) service personnel. Panel position above the floor must be per National Electric Code (NEC) 2002 Article 404.8 or NEC 1999 Article 380-8 or other national or local code.

A recommended method for providing a Main Disconnect Control with multi-point remote control capability is shown in Illustration 5-1.

5-3-1 Main Disconnect Control (MDC) (Continued)

Time delay fuses are recommended on the lines providing power to the protective disconnect device and must be rated according to the cable size. The circuit breaker ahead of the fuses must be capable of handling the magnetizing inrush current of the Shield Cooler Compressor and transformer of the PDU module (PD1) in the ACGD/PDU Cabinet (MR3).

Check local and national codes to determine if an interlock to the air-conditioning unit in the Computer/Equipment Room is required in the protective disconnect set-up.

Customer supplied emergency off buttons are to be mounted in the Magnet Room near each exit 60 in. (1524 mm) from the floor and connected to the protective disconnect device in order to disable the power to the PDU in emergency situations. This button is to be clearly labeled "Emergency Off" and visible to personnel in the Magnet Room. It is important that the button be labeled "off" and not "stop" since there exists an "Emergency Stop" button in the Signa system which powers down different equipment than the emergency off button installed by the facility. An additional emergency off button should be mounted in the computer/equipment room near the exit door.

Customer supplied main disconnect controller (MDC) must be lockable to meet OSHA requirements for power Lockout/Tagout requirements. An optional MDC is available from GE providing for the disconnect of the facility power to the PDU, including the emergency off buttons, contacts for an interlock to the air-conditioning units in the computer/equipment room, and the Shield Cooler Compressor Cabinet power circuit with auto restart and emergency off functions.

The Main Disconnect Control Panel must be listed and labeled by a Nationally Recognized Testing Lab (NRTL) such as Underwriters Laboratory (UL) in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110-2.

Note

The maximum conductor the GE available option MDC can accept is #3/0 AWG (95 mm²). For feeders larger than 3/0 AWG (83 mm²) the wires must be reduced (ie. splice, junction box, etc.) to 3/0 AWG (83 mm²) within 10 feet (3 meters) of MDC. It is important to note the maximum cable wire from the MDC to the PDU must not be larger than the maximum wire the PDU will accept, refer to Section 5-3-2 System Power Distribution Unit.

■ 5-3-2 System Power Distribution Unit

The PDU Module in the lower portion of ACGD/PDU Cabinet has an integrated filter for a level of power conditioning.

■ The largest allowable phase conductor the PDU will accept is 2/0 AWG (70 mm²). Larger feeder wires can be connected to the MDC with 2/0 AWG (70 mm²) between the MDC and PDU.

Note

The ground conductor shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.

Note

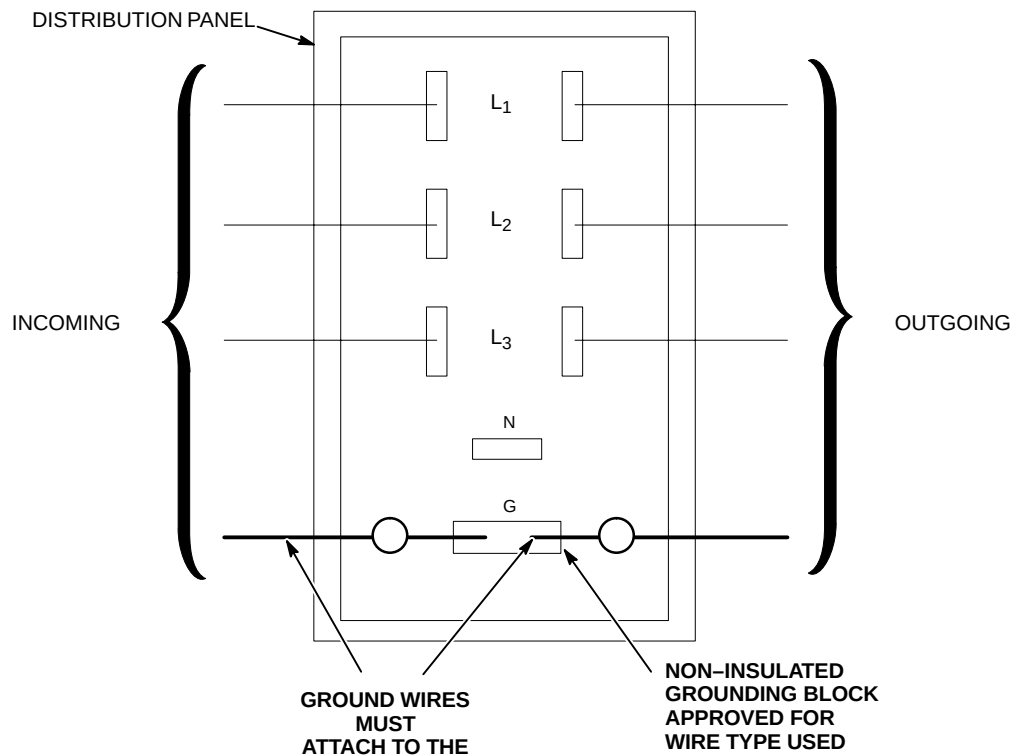
Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of ACGD/PDU Cabinet (MR3).

5-4 GROUNDING

5-4-1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into a facility, and be continuous to the MR system power disconnect in the room. This ground can be spliced with "High Compression Fittings" and should be terminated at each distribution panel it passes through. When the ground is broken for a connection to a panel, the ground shall be connected into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the grounding block. See Illustration 5-2.

The connection at the power source shall be at the grounding point of the "Neutral – Ground" if a "Wye" transformer is used, or the grounded phase of a grounded "Delta", or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.



GROUND CONNECTION AT DISTRIBUTION PANEL

ILLUSTRATION 5-2

M4301A

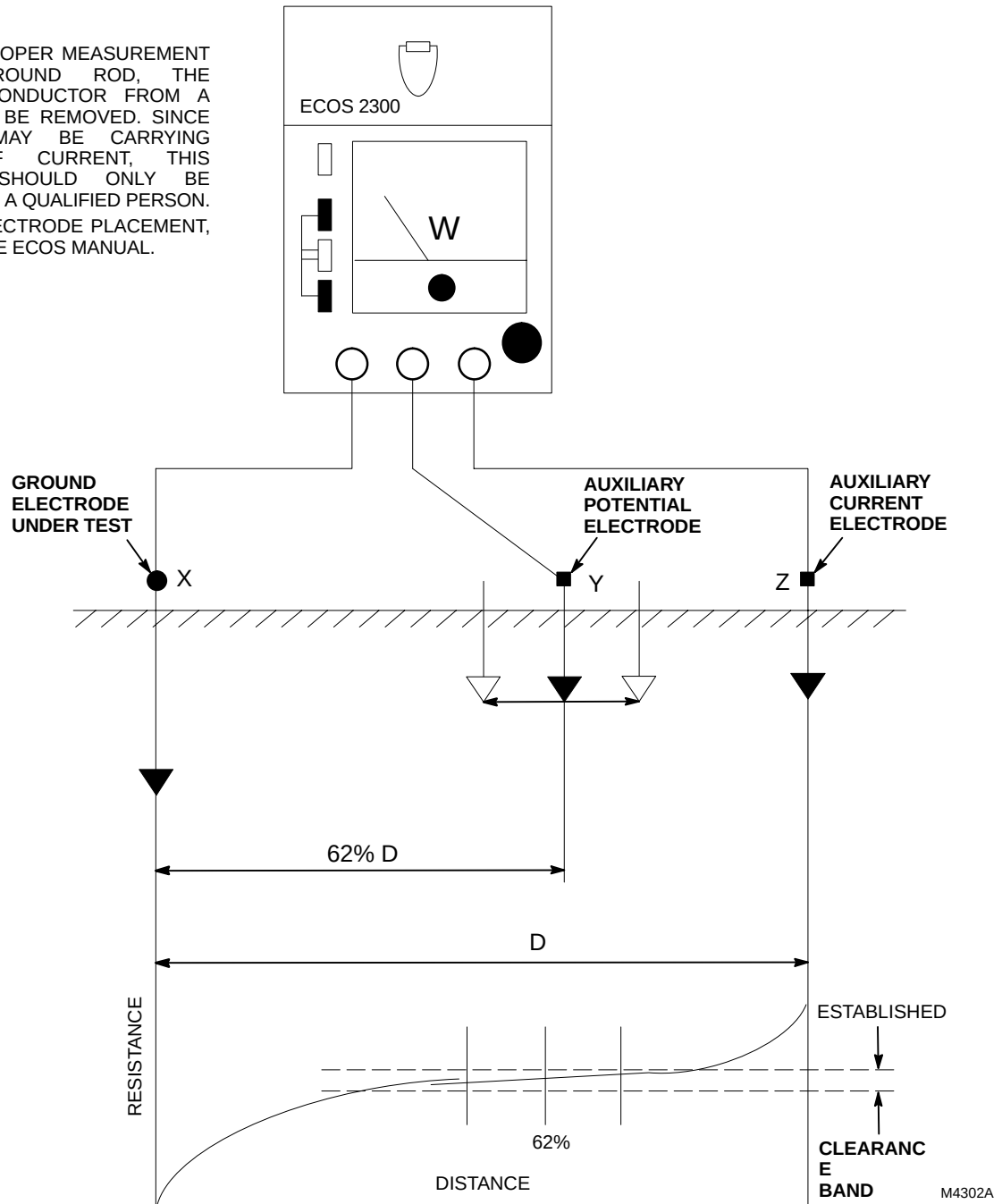
Ground Wire

The ground wire shall be copper wire with a minimum of AWG 1/0 or the same size as the power feeders whichever is larger. This means that if there is a primary feeder to a distribution panel of 500 MCM with a secondary feeder to the MR system of AWG 2 wire, the ground to the distribution panel shall be 500 MCM with an AWG 1/0 to the MR system. The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 – 1982. Refer to Illustration 5-3 for typical equipment and methods to measure the different portions of the 2 ohm impedance.

5-4-1 Facility Ground (Continued)

NOTE:

- TO ENSURE PROPER MEASUREMENT OF THE GROUND ROD, THE GROUNDING CONDUCTOR FROM A FACILITY MUST BE REMOVED. SINCE THIS WIRE MAY BE CARRYING AMOUNTS OF CURRENT, THIS PROCEDURE SHOULD ONLY BE PERFORMED BY A QUALIFIED PERSON.
- DETAILS OF ELECTRODE PLACEMENT, ETC., ARE IN THE ECOS MANUAL.



STANDARD HOOK-UP TO SYSTEM UNDER TEST FOR GROUND ROD IMPEDANCE
ILLUSTRATION 5-3

■ 5-4-2 System Ground

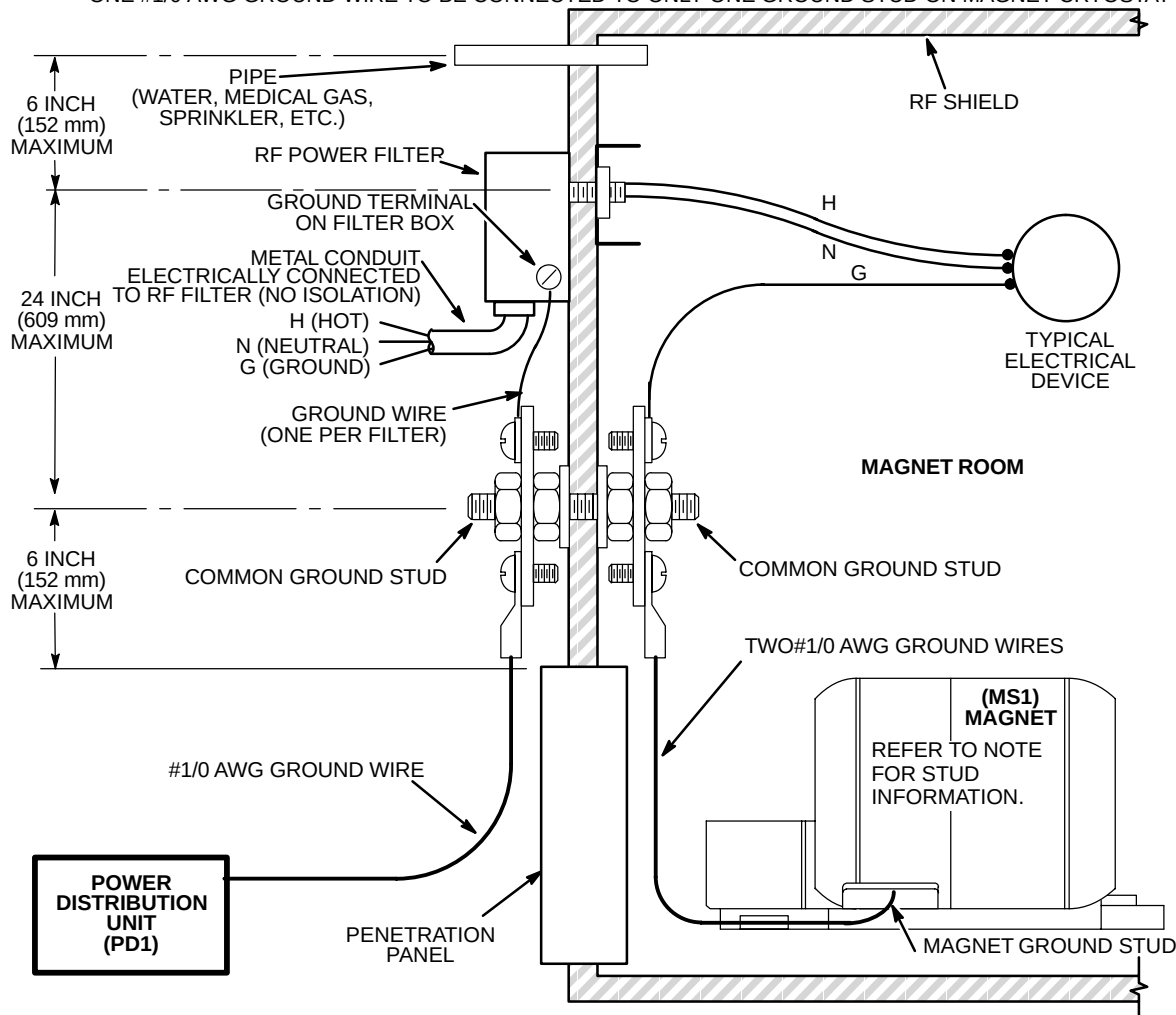
The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF Shielded Room common ground point. **This RF Shielded Room common ground point is to be located within 6 in. (152 mm) of the GE supplied penetration panel.** Refer to Section 7, RF SHIELDED ROOM, for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in Illustration 5-4 must be met when running power lines into the Magnet Room.

Any modifications or non-MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

5-4-2 System Ground (Continued)

- NOTE:**
- ALL ITEMS SHOWN ARE CUSTOMER SUPPLIED EXCEPT POWER DISTRIBUTION UNIT, MAGNET, AND TWO #1/0 AWG GROUND WIRES BETWEEN MAGNET GROUND STUD AND RF COMMON GROUND POINT.
 - **RESISTANCE BETWEEN ANY TWO GROUNDED DEVICES MUST NOT EXCEED 0.1 OHM TO ENSURE EQUAL POTENTIAL GROUND SYSTEM WITHIN MAGNET ROOM.**
 - **LOCATE FILTERS WITHIN 2 FEET (600 mm) OF RF COMMON GROUND STUD WHICH MUST BE LOCATED WITHIN 6 INCHES (152 mm) OF PENETRATION PANEL.**
 - ALL EXTERNAL CONDUIT MUST BE METAL AND ELECTRICALLY CONNECTED TO THE RF POWER FILTERS (IE. NO ISOLATION) UNLESS THE FILTERS ARE LOW VOLTAGE (<30 VOLTS).
 - RF POWER FILTERS OF 30 VOLTS OR LESS MAY BE LOCATED ANYWHERE ON THE RF SHIELD **PROVIDED THE INCOMING CONDUIT IS NON-METALLIC OR IS DIELECTRICALLY ISOLATED AND WITH NO GROUND WIRE. IF THE INCOMING CONDUIT IS METALLIC AND/OR HAVE A GROUND WIRE, THESE FILTERS MUST ALSO BE LOCATED WITHIN 24 INCHES (609 mm) OF THE RF COMMON GROUND STUD.**
 - ALL CONDUITS IN THE RF ROOM MUST BE METAL. STEEL IS ACCEPTABLE PROVIDED IT IS ADEQUATELY ANCHORED.
 - ALL ELECTRICAL DEVICES (IE. OUTLETS, LIGHT FIXTURES, ETC.) MUST HAVE A GROUND WIRE FROM ITS POWER SOURCE AND BE GROUNDED TO RF ROOM SHIELD AT THE RF COMMON GROUND STUD AS SHOWN BELOW.
 - ALL METALLIC PIPES ENTERING THE RF ROOM, EXCLUDING CRYOGENIC VENT AND FLOOR DRAINS, MUST BE LOCATED WITHIN 30 INCHES (762 mm) OF THE RF COMMON GROUND.
 - ONE #1/0 AWG GROUND WIRE TO BE CONNECTED TO ONLY ONE GROUND STUD ON MAGNET CRYOSTAT OR FOOT.



MR MAGNET ROOM GROUNDING REQUIREMENTS AND TYPICAL DIAGRAM

ILLUSTRATION 5-4

5-5 GROUND FAULT PROTECTION

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2002 Article 517.2 definitions or NEC1999 Article 517–3 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC 2002 Article 517.17 or NEC 1999 Article 517–17. NEC 2002 Article 517.17(A) or NEC 1999 Article 517–17(a) states “Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load.”

NEC 2002 Article 230.95 or 215.10 or NEC 1999 Article 230–95 or 215–10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2002 Article 517.17(B) or NEC 1999 Article 517–17(b).

Check national and local electrical codes.